

SMART FARMER
SMART SPARE PARTS MANAGEMENT SYSTEM FOR
AGRICULTURE MACHINERY

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ABSTRACT

Agriculture in Sri Lanka and many developing regions is highly dependent on the reliability and continuous operation of mechanical equipment such as tractors, tillers, and irrigation systems. The performance and longevity of these machines are directly influenced by the condition, authenticity, and management of their spare parts. However, current practices for spare part identification, wear assessment, lifecycle management, and verification remain largely manual, fragmented, and reactive. These limitations lead to frequent equipment failures, increased maintenance costs, and the widespread circulation of counterfeit or low-quality components. The Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System is proposed as a comprehensive, multi-component solution to address these challenges through the integration of artificial intelligence, computer vision, blockchain technology, and data-driven predictive analytics. The system is composed of four key components: an Image Recognition and 3D Visualization module for intuitive spare part identification, a Computer Vision Wear Detection module for automated defect analysis, a Lifecycle Forecasting module for predicting remaining useful life, and a Blockchain-Based Verification and Traceability module to ensure authenticity and secure ownership tracking. The Computer Vision Wear Detection module, serving as the perceptual intelligence layer, evolved through a two-phase approach. An initial MobileNetV2-based classification model demonstrated reasonable accuracy in identifying worn components but lacked spatial awareness. This limitation was addressed by adopting YOLOv8, a real-time object detection framework capable of precisely localizing multiple defects such as V-belt fraying and tractor tyre cracking. The model was trained on real-world datasets collected from farms and engineering workshops, ensuring robustness under practical conditions. In parallel, the Image Recognition and Visualization module employs deep learning techniques to identify spare parts and map them to interactive 3D representations, improving user understanding of component structure and placement. The Lifecycle Forecasting module integrates wear detection outputs and usage data to estimate remaining useful life and support predictive maintenance decisions. Complementing these, the Blockchain-Based Verification module leverages a permissioned Hyperledger Fabric network to provide immutable spare part registration, QR-based authentication, and secure ownership transfer, thereby addressing issues of counterfeit components and lack of traceability. Experimental evaluations demonstrate that the integrated system enhances detection accuracy, improves maintenance decision-making, and ensures secure and transparent spare part management. By combining multiple advanced technologies into a unified platform, this research contributes a practical and scalable solution for transforming agricultural machinery maintenance from a reactive process into a proactive, intelligent system.

Keywords: Smart agriculture, computer vision, YOLOv8, predictive maintenance, spare part lifecycle, blockchain, traceability, deep learning

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BPCAT	Blockchain-based Product-centred Confidentiality-preserving Architecture for Traceability
CI/CD	Continuous Integration / Continuous Deployment
CNN	Convolutional Neural Network
DeFi	Decentralized Finance
FYP	Final Year Project
IPFS	InterPlanetary File System
IoT	Internet of Things
LKR	Sri Lankan Rupee
MRO	Maintenance, Repair, and Overhaul
NFC	Near-Field Communication
OEM	Original Equipment Manufacturer
PII	Personally Identifiable Information
REST	Representational State Transfer
RFID	Radio-Frequency Identification
SDK	Software Development Kit
SRS	Software Requirement Specification
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
VM	Virtual Machine
WBS	Work Breakdown Structure
ZKP	Zero-Knowledge Proof

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Agriculture continues to play a vital role in the economic and social structure of Sri Lanka, particularly in rural regions where a significant portion of the population depends on farming as their primary source of income. The sector encompasses a wide range of activities including paddy cultivation, plantation crops such as tea and rubber, and small-scale vegetable farming. Over the past two decades, there has been a noticeable transition from traditional labour-intensive practices to **mechanised farming**, driven by the need to increase productivity, reduce labour costs, and meet growing food demands.

Machinery such as tractors, power tillers, irrigation pumps, and harvesting equipment has become an integral part of modern agricultural operations. However, the efficiency and reliability of these machines are heavily dependent on the **condition, availability, and authenticity of their spare parts**. Components such as V-belts, tractor tyres, bearings, and transmission elements are subjected to continuous mechanical stress, environmental exposure, and operational wear. As a result, they require timely maintenance and replacement to ensure uninterrupted machine performance.

Despite the increasing reliance on machinery, spare part management practices in Sri Lanka and similar developing agricultural economies remain largely **informal, reactive, and unstructured**. Farmers typically rely on manual inspection, prior experience, or advice from local mechanics to assess part condition. This approach suffers from several limitations, including inconsistency, lack of precision, and dependency on individual expertise. In many cases, maintenance decisions are made only after a component fails, leading to unexpected machine downtime, increased repair costs, and potential crop losses.

Another major challenge is the widespread circulation of **counterfeit and substandard spare parts** within local supply chains. Farmers often lack reliable mechanisms to verify the authenticity of purchased components, resulting in reduced equipment lifespan and compromised safety. Additionally, there is no standardized system for tracking spare part ownership, usage history, or lifecycle events, making it difficult to maintain accurate records or make data-driven decisions.

The global shift toward **precision agriculture and smart farming** introduces new opportunities to address these challenges. Technologies such as artificial intelligence (AI), computer vision, Internet of Things (IoT), and blockchain have demonstrated significant potential in transforming traditional agricultural practices into **intelligent, data-driven systems**. These technologies enable real-time monitoring, predictive maintenance, automation, and secure data management.

In this context, the **Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System** is proposed as a comprehensive solution to modernize spare part management in agriculture. The system integrates multiple advanced technologies to provide a unified platform capable of identifying spare parts, detecting wear conditions, predicting lifecycle, and ensuring secure verification and traceability.

1.1.1 System Components Overview

The proposed system is designed as a **multi-component architecture**, where each module addresses a specific challenge within the spare part management lifecycle. The four main components are described below:

a. Image Recognition and 3D Visualization Module

This component focuses on the **identification and understanding of spare parts** using deep learning techniques. Traditional identification methods rely heavily on user knowledge, which is often limited among smallholder farmers. This module utilizes convolutional neural networks to classify spare parts based on input images and maps the identified components to corresponding **3D models**.

The integration of 3D visualization provides an intuitive representation of machinery, allowing users to:

- Understand the spatial placement of components
- Explore machine structure interactively
- Gain contextual knowledge of part functionality

This enhances usability and bridges the knowledge gap between technical systems and end users.

b. Computer Vision Wear Detection Module

The Computer Vision Wear Detection Module serves as the **perceptual intelligence layer** of the system. Its primary function is to analyze images of spare parts and automatically detect wear and damage.

The research follows a two-phase evolution:

- Phase One: MobileNetV2-based classification for detecting overall wear condition
- Phase Two: YOLOv8-based object detection for precise localization of defects

This module identifies defects such as:

- Cracks and surface damage
- Fraying and exposed threads (V-belts)
- Bald patches and sidewall cracks (tyres)

By providing **spatially-aware detection**, the module enables accurate assessment of wear severity, which is critical for predictive maintenance.

c. Spare Part Lifecycle Forecasting Module

This component introduces **predictive intelligence** into the system. Instead of relying on reactive maintenance, it uses machine learning techniques to estimate the **remaining useful life (RUL)** of spare parts.

The module integrates:

- Wear detection outputs (defect type, severity)
- Usage data and operational patterns

By analysing these factors, the system can:

- Predict failure timelines
- Recommend maintenance schedules
- Optimize spare part replacement decisions

This transforms maintenance from a reactive process into a **proactive, data-driven strategy**.

d. Blockchain-Based Verification and Traceability Module

The Blockchain module acts as the **trust and security layer** of the system. It addresses the issue of counterfeit spare parts and lack of transparency in supply chains.

Using a permissioned blockchain framework (Hyperledger Fabric), the system:

- Assigns unique digital identities to spare parts
- Records transactions in an immutable ledger
- Enables QR-based authentication for real-time verification
- Tracks ownership and lifecycle events

This ensures:

- Data integrity
- Transparency
- Secure and tamper-proof records

1.2 Problem Statement

Agricultural spare part management in Sri Lanka is characterized by **lack of automation, limited data availability, and absence of reliable verification mechanisms**. These limitations result in inefficient maintenance practices and increased operational risks.

Manual inspection methods are:

- Subjective and inconsistent
- Lacking spatial precision
- Dependent on expert knowledge

Furthermore, spare part supply chains suffer from:

- Poor traceability
- Fragmented ownership records
- Presence of counterfeit components

Existing technological solutions are often **isolated**, addressing only a single aspect such as detection or verification, without providing a complete system.

1.2.1 Component-wise Problem Context

a. Image Recognition and Visualization

Users face difficulty in identifying spare parts and understanding their location within machinery due to lack of intuitive tools.

b. Wear Detection

Existing approaches cannot accurately detect and localize wear defects, limiting their practical usefulness.

c. Lifecycle Forecasting

There is no system to predict spare part failure or remaining useful life based on real-world data.

d. Blockchain Verification

There is no secure, decentralized mechanism for verifying authenticity and tracking spare part lifecycle.

1.3 Research Objectives

1.3.1 Main Objective

To develop an integrated intelligent system for spare part identification, wear detection, lifecycle forecasting, and secure verification in agricultural machinery.

1.3.2 Specific Objectives

a. Image Recognition and Visualization

- Develop a deep learning model for spare part identification
- Integrate 3D visualization for enhanced understanding

b. Wear Detection

- Develop a YOLOv8-based defect detection model
- Enable spatial localization and multi-defect detection

c. Lifecycle Forecasting

- Predict remaining useful life (RUL)
- Support proactive maintenance decisions

d. Blockchain Verification

- Implement blockchain-based verification system
- Enable QR-based authentication and ownership tracking

1.4 Scope of the Research

The research is limited to:

- Selected spare parts (e.g., tyres, belts)
- Real-world datasets collected in Sri Lanka
- Mobile-based deployment for accessibility
- Integration of AI and blockchain technologies

1.5 Justification and Significance

This research is significant in addressing both **practical agricultural challenges and technological gaps**.

Practically, it:

- Reduces machinery downtime
- Improves maintenance efficiency
- Minimizes economic losses for farmers

Technologically, it:

- Demonstrates integration of multiple advanced systems
- Provides a real-world dataset
- Contributes to smart agriculture research

Ultimately, the system supports the transition toward **accessible precision agriculture**, enabling smallholder farmers to benefit from advanced technologies.

CHAPTER 2: METHODOLOGY

2.1 Overview of the Research Methodology

The methodology of the **Smart Farmer AI Integrated Spare Part Lifecycle Forecasting System** was designed as a **multi-component, modular, and integrated system development approach**. Since the proposed system addresses several problems in agricultural spare part management, the methodology was divided into four major component workflows:

- a. Spare Part Image Recognition and 3D Visualization System
- b. Computer Vision Wear Detection Module
- c. Dynamic Compatibility Recommendation System
- d. Blockchain-Based Spare Parts Verification and Traceability System

Each component was developed using a separate technical methodology, but all components were designed to operate as part of one unified Smart Farmer platform. The overall methodology includes real-world data collection, dataset preparation, artificial intelligence model development, recommendation logic design, blockchain-based verification, backend integration, and final system evaluation.

The general workflow of the system can be summarized as follows:

1. Identify agricultural spare part management problems through domain analysis.
2. Collect real-world data from farms, workshops, spare part shops, farmers, vendors, and technical users.
3. Prepare datasets for image recognition, wear detection, compatibility recommendation, and verification.
4. Develop AI models for spare part identification and defect detection.
5. Develop a compatibility-aware recommender system for spare part substitution.
6. Implement blockchain-based spare part authentication and lifecycle tracking.
7. Integrate all components through backend APIs and user interfaces.
8. Evaluate each component individually and validate the complete system workflow.

This methodology ensures that the system is not only technically functional but also practical for real-world agricultural environments.

2.2 Component A: Spare Part Image Recognition and 3D Visualization System

2.2.1 Methodological Approach

The methodology of the Image Recognition and 3D Visualization component follows a **deep learning-based classification and visualization pipeline**. The purpose of this component is to identify agricultural machinery spare parts from images and provide users with a visual understanding of where the identified component is located within the machine.

Unlike a basic image classification system, this component does not stop after predicting the spare part name. Instead, the classified spare part is mapped to a corresponding 3D machine model, where the part is highlighted for better user understanding. This makes the system useful for farmers, technicians, trainees, and spare part sellers who may not have expert mechanical knowledge.

The main stages of this component are:

1. Target spare part selection
 2. Image dataset collection
 3. Data preprocessing and augmentation
 4. ResNet-50 model development
 5. Model training and evaluation
 6. 3D model preparation
 7. Mapping recognition output to 3D visualization
 8. Integration with web/mobile interface
-

2.2.2 Target Spare Part Selection

The first step was to identify spare parts that are commonly used in agricultural machinery and are suitable for visual recognition. Selected components included parts such as clutch plates, pressure plates, filters, shafts, bearings, gears, axles, and other machine components.

The selection was based on:

- Availability of parts in local spare part shops
- Relevance to agricultural machinery maintenance
- Visual distinguishability of the parts
- Practical usefulness for farmers and mechanics

This ensured that the component focused on real spare parts that users are likely to encounter in practical maintenance scenarios.

2.2.3 Data Collection

A primary dataset was created because publicly available datasets for Sri Lankan agricultural machinery spare parts are limited. Images were collected from:

- Local spare part shops
- Agricultural machinery workshops
- Catalogues
- Real-world environments where spare parts are stored or sold

Images were captured under different real-world conditions, including varying lighting, angles, distances, backgrounds, and smartphone camera quality. This was important because the final system is expected to be used by ordinary users in practical environments rather than controlled laboratory conditions.

The dataset included multiple images per spare part category. According to the individual component document, the image recognition dataset included **902 images** across multiple spare part classes.

2.2.4 Data Preprocessing

After data collection, the images were cleaned and standardized before model training. The preprocessing process included:

- Removing blurred images
- Removing overexposed or underexposed images
- Removing irrelevant images
- Removing near-duplicate images
- Resizing images to **224 × 224 pixels**
- Converting images into RGB format
- Normalizing pixel values to improve training stability

This step ensured that the dataset was clean, consistent, and suitable for deep learning training.

2.2.5 Annotation Methodology

Since this component performs image classification rather than object detection, annotation was done at the image level. Each image was assigned a single class label representing the spare part category.

For example:

- Pressure Plate
- Clutch Plate
- Bearing Release
- Oil Filter
- Fuel Filter
- Shaft PTO
- Water Pump Assembly

A verification step was also used to ensure that labels were correctly assigned. Incorrect labels were corrected before training to prevent model confusion.

2.2.6 Data Augmentation

Data augmentation was applied to increase dataset diversity and reduce overfitting. Since real-world users may capture spare part images from different angles and lighting conditions, augmentation helped simulate these variations.

The augmentation techniques included:

- Random rotation
- Horizontal flipping
- Brightness adjustment
- Contrast adjustment
- Random zoom
- Positional shifting

These transformations helped the model generalize better when exposed to unseen images.

2.2.7 ResNet-50 Model Development

The component used **ResNet-50** as the main deep learning model. ResNet-50 was selected because it is a strong convolutional neural network architecture that can learn complex visual features while avoiding the vanishing gradient problem through residual connections.

The model was developed using transfer learning. Instead of training from scratch, pre-trained ImageNet weights were used as the starting point. This allowed the model to reuse general visual features such as edges, shapes, and textures, while learning domain-specific spare part features during fine-tuning.

The model development process included:

1. Loading the ResNet-50 base model with pre-trained weights.
 2. Removing the original classification head.
 3. Adding a new classification head for spare part classes.
 4. Freezing base layers during initial training.
 5. Training the new classification layers.
 6. Unfreezing selected top layers.
 7. Fine-tuning the model with a lower learning rate.
-

2.2.8 Model Training Configuration

The training configuration included:

- Input size: $224 \times 224 \times 3$
- Optimizer: Adam
- Loss function: Categorical Cross-Entropy
- Evaluation metric: Accuracy
- Dropout layer to reduce overfitting
- Early stopping based on validation loss
- Model checkpointing to save the best model

The model was trained in a GPU-supported environment and evaluated using training, validation, and test datasets.

2.2.9 Inference Pipeline

During inference, the user uploads or captures an image of a spare part. The system performs the following steps:

1. Resize the input image to 224×224 pixels.
2. Normalize the image.
3. Pass the image through the trained ResNet-50 model.
4. Generate probability scores for all spare part classes.
5. Select the class with the highest probability.
6. Return the predicted spare part name and confidence score.
7. Retrieve corresponding spare part details from the database.

The output of this stage becomes the input for the 3D visualization process.

2.2.10 3D Visualization Methodology

After identifying the spare part, the system maps the predicted part to a corresponding component in a 3D model of the machine. The purpose is to help users understand not only what the part is, but also where it is located and how it relates to other parts.

The 3D visualization process includes:

- Preparing a digital 3D model of the machine
- Separating important components as identifiable objects
- Assigning each spare part class to a 3D object ID
- Highlighting the detected component in the 3D model
- Allowing user interaction such as rotation, zooming, and navigation

- Providing visual explanations or AI-generated videos where applicable

This improves practical usability because users can visually understand the part location without relying only on text output.

2.2.11 Deployment Pathway

For deployment, the trained model can be converted into a lightweight format such as TensorFlow Lite. This enables efficient inference on mobile or web-based platforms. The classification output is returned through an API and connected to the visualization layer.

The final output of Component A includes:

- Spare part name
- Confidence score
- Relevant part details
- Highlighted 3D component
- Visual explanation for user understanding

2.3 Component B: Computer Vision Wear Detection Module

2.3.1 Methodological Approach

The Computer Vision Wear Detection component follows a **two-phase experimental methodology**. The purpose of this component is to detect and localize wear defects in agricultural spare parts, especially V-belts and tractor tyres.

The first phase used MobileNetV2 as a lightweight classification model. However, classification only provides a whole-image label and cannot identify the exact location of damage. Therefore, the second phase introduced YOLOv8, an object detection model capable of locating defects using bounding boxes.

The methodology includes:

1. Identification of target spare parts and wear classes
2. Real-world image data collection
3. Image preprocessing and annotation
4. Phase One: MobileNetV2 classification
5. Identification of classification limitations
6. Phase Two: YOLOv8 object detection
7. Model training and hyperparameter configuration
8. Performance evaluation
9. Deployment preparation

This component acts as the **perceptual intelligence layer** of the system by converting raw images into structured wear condition data.

2.3.2 Target Spare Parts and Wear Classes

The component focuses on two important agricultural spare part categories:

- V-belts
- Tractor tyres

These parts were selected because they are commonly used in agricultural machinery and are highly affected by wear.

The defect classes include:

For V-belts:

- Healthy / No defect
- Surface crack / glazing
- Exposed thread / fraying

For tractor tyres:

- Healthy / No defect
- Bald patch / smooth tread
- Sidewall crack
- Lug wear / tread loss

These classes were selected based on common real-world wear patterns observed in farms and workshops.

2.3.3 Data Collection from Farms

Images were collected from local farms where V-belts and tractor tyres were still in use. This allowed the dataset to capture real operational conditions such as:

- Dirt and mud
- Uneven lighting
- Outdoor shadows
- Different camera angles
- Parts still attached to machines

Collecting farm images was important because the final users will capture images using smartphones in similar environments. Therefore, the training data had to reflect deployment conditions.

2.3.4 Data Collection from Workshops

In addition to farms, images were collected from local iron and engineering workshops. Workshops provided access to removed and damaged parts, including advanced wear cases that may not frequently appear during farm visits.

Workshop images included:

- Severely cracked V-belts
- Frayed belts
- Bald tyres
- Sidewall cracks
- Broken or heavily worn tread sections

This improved class balance and ensured that the model learned severe defect patterns.

The individual wear detection methodology states that the raw dataset included around **1,220 images**, later reduced to a working set of **1,047 images** after quality screening.

2.3.5 Image Quality Screening

All collected images were manually screened before training. Images were removed if they had:

- Severe blur
- Poor focus
- Extreme overexposure
- Extreme underexposure
- Defects not clearly visible
- Parts occupying too small an area
- Near-duplicate content

This step was essential because object detection performance depends heavily on image and annotation quality.

2.3.6 Annotation Methodology

Two types of annotation were used:

Phase One Annotation

For MobileNetV2 classification, each image was assigned one label representing the overall wear class.

Phase Two Annotation

For YOLOv8 object detection, bounding boxes were drawn around each visible defect region using annotation tools such as LabelImg.

Bounding boxes were drawn around:

- Exposed thread regions
- Surface cracks
- Bald patches
- Sidewall cracks
- Tread loss areas

Each box included:

- Class label
- Coordinates
- Defect region boundary

This allowed YOLOv8 to learn both defect type and location.

2.3.7 Annotation Quality Control

To maintain annotation consistency, clear annotation rules were followed. Defects were boxed tightly around the visible damage area. Where multiple defects existed in one image, each defect was annotated separately.

Annotation quality was checked using visual inspection and Intersection over Union (IoU)-based agreement where applicable. This was important because inaccurate bounding boxes can reduce detection quality.

2.3.8 Dataset Splitting

The dataset was divided into:

- Training set: 70%
- Validation set: 15%
- Test set: 15%

Stratified splitting was used to maintain class distribution across all subsets. This prevented rare classes from being underrepresented during evaluation.

2.3.9 Data Augmentation

Augmentation was applied to improve generalization. The following transformations were used:

- Horizontal flip

- Random rotation
- Brightness variation
- Contrast variation
- Random zoom
- Gaussian noise
- Positional shifting
- Mosaic augmentation for YOLOv8
- HSV color augmentation

For object detection, bounding box coordinates were transformed together with the image. This preserved annotation correctness.

2.3.10 Phase One: MobileNetV2 Classification

MobileNetV2 was used as the baseline model due to its lightweight architecture and suitability for mobile deployment.

The training process included:

1. Loading MobileNetV2 with ImageNet weights.
2. Removing the original classification layer.
3. Adding a new classification head.
4. Freezing the base model.
5. Training the classification head.
6. Fine-tuning top layers.

Training configuration included:

- Input size: 224×224
 - Optimizer: Adam
 - Loss function: Categorical Cross-Entropy
 - Learning rate: $1e-4$ initially, reduced during fine-tuning
 - Early stopping
 - Class weighting for imbalance
-

2.3.11 Limitations Identified in MobileNetV2

Although MobileNetV2 achieved reasonable classification performance, it had three major limitations:

1. It could not localize defects.
2. It could not detect multiple defects in one image.
3. It could not calculate severity from defect size.

For example, if a tyre had both a bald patch and a sidewall crack, the classifier could only return one label. This made it unsuitable for lifecycle forecasting, because forecasting requires spatial and severity information.

2.3.12 Phase Two: YOLOv8 Object Detection

YOLOv8 was selected to overcome the limitations of classification. Unlike MobileNetV2, YOLOv8 detects individual defects and returns bounding boxes.

YOLOv8 provides:

- Defect class
- Bounding box coordinates
- Confidence score
- Multi-defect detection
- Spatial defect location
- Severity estimation from bounding box area

YOLOv8s was selected because it provides a good balance between detection accuracy and mobile deployment efficiency.

2.3.13 YOLOv8 Training Configuration

The YOLOv8 model was trained using:

- Model variant: YOLOv8s
- Input size: 640×640
- Epochs: 100
- Batch size: 16
- Optimizer: SGD
- Learning rate: 0.01
- Momentum: 0.937
- Weight decay: 0.0005
- IoU threshold: 0.45
- Confidence threshold: 0.5 during evaluation

The model used classification loss, bounding box regression loss, and object confidence loss.

2.3.14 YOLOv8 Output Processing

The output of YOLOv8 includes multiple detection objects. Each detection includes:

- Defect class
- Confidence score
- Bounding box coordinates
- Defect area
- Derived severity score

The severity score is calculated based on the size of the bounding box relative to the image or part area. This allows the system to pass meaningful numerical wear data to the lifecycle forecasting module.

2.3.15 Model Evaluation

MobileNetV2 was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

YOLOv8 was evaluated using:

- Precision
- Recall
- F1-score
- mAP50
- Per-class AP
- Qualitative detection examples

The YOLOv8 model achieved stronger practical value because it provided localization and multi-defect detection, which classification could not provide.

2.3.16 Deployment Pathway

For deployment, the YOLOv8 model was converted from PyTorch format to ONNX and then to TensorFlow Lite. Quantization was applied to reduce model size and improve mobile performance.

The final output of Component B includes:

- Detected defect type
- Bounding box location
- Confidence score
- Severity score
- Structured JSON response for forecasting and UI display

2.4 Component C: Dynamic Compatibility Recommendation System

2.4.1 Methodological Approach

The Dynamic Compatibility Recommendation System follows a **hybrid recommendation methodology**. Its purpose is to recommend compatible and functionally equivalent spare parts when original parts are unavailable, expensive, or delayed.

Unlike ordinary recommendation systems that rely only on item similarity or user behavior, this component must consider mechanical compatibility. Therefore, the methodology combines:

- Structured compatibility rules
- Semantic similarity
- Random Forest compatibility prediction
- Feedback-aware ranking
- Explanation generation

This component is designed to recommend parts that are not only textually similar but also mechanically and functionally suitable.

2.4.2 Data Collection and Preparation

The data required for this component includes:

- Spare part name
- Brand
- Category
- Machine model
- Description
- Specifications
- Diameter

- Material
- Price
- Expected lifespan
- Machine family
- Compatibility group
- Function type
- Substitute level
- Confidence score
- Evidence source

The data was collected from available spare part records, domain knowledge, technical specifications, vendor inputs, and compatibility relationships.

Data preparation included:

- Removing duplicate records
- Standardizing machine model names
- Correcting category inconsistencies
- Validating numeric fields
- Handling missing values
- Converting specifications into structured JSON format

This step was important because incorrect data can lead to unsafe recommendations.

2.4.3 Compatibility Schema Design

A structured compatibility schema was designed to represent domain-specific mechanical relationships. The schema includes:

- **Machine family:** Groups machines with similar operating characteristics.
- **Compatibility group:** Links interchangeable or comparable parts.
- **Function type:** Defines the mechanical role of the part.
- **Substitute level:** Indicates exact, near, or functional substitute.
- **Confidence score:** Represents recommendation reliability.
- **Evidence source:** Indicates whether compatibility is based on manufacturer data, expert input, field feedback, or system inference.

This schema allows the recommender to go beyond keyword similarity and evaluate actual compatibility.

2.4.4 Rule-Based Filtering

Rule-based filtering was used as the safety layer of the recommender.

Hard Rules

Hard rules eliminate unsafe candidates before ranking. These include:

- Candidate must belong to the same or compatible category.
- Candidate must have the same function type.
- Candidate must belong to a compatible machine family.
- Candidate must have a valid compatibility group.
- Original selected part must be excluded from recommendations.

Hard rules ensure that mechanically invalid parts are not recommended even if they have similar names.

Soft Rules

Soft rules influence ranking but do not automatically remove candidates. These include:

- Diameter closeness
- Material similarity
- Price difference
- Lifespan difference
- Feedback acceptance
- Substitute level

Soft rules allow flexibility while still improving recommendation quality.

2.4.5 Similarity-Based Ranking

After hard filtering, vector-based similarity ranking is applied. Spare part records are converted into vector representations using fields such as:

- Name
- Brand
- Category
- Machine model
- Description
- Specifications

Similarity scores are calculated to identify candidates that are semantically close to the selected part.

This is useful because farmers and vendors may use different terms for the same or similar spare parts. However, similarity is applied only after rule filtering to prevent invalid matches.

2.4.6 Pairwise Dataset Construction

For machine learning prediction, the data is converted into pairwise format. Each record contains:

- Part A
- Part B
- Compatibility label

Positive pairs are created from known compatible or substitute parts. Negative pairs are created from parts with different functions, incompatible categories, or unrelated machine families.

This pairwise dataset allows the model to learn compatibility relationships between two parts rather than classifying a single item.

2.4.7 Feature Engineering

The Random Forest model uses binary and numeric features.

Binary features include:

- Same category
- Same machine family
- Same compatibility group
- Same function type
- Same material

Numeric features include:

- Diameter difference
- Price difference
- Lifespan difference
- Similarity score

These features reflect mechanical, semantic, and economic compatibility.

2.4.8 Random Forest Compatibility Prediction

A Random Forest classifier is trained using the pairwise dataset. Random Forest was selected because it performs well on tabular data, handles non-linear relationships, and is more interpretable than many black-box models.

The model predicts the probability that two spare parts are compatible.

The training process includes:

1. Split pairwise dataset into training and testing data.
2. Train Random Forest classifier.
3. Evaluate using classification metrics.
4. Save trained model.
5. Integrate model into backend recommendation service.

Evaluation metrics include:

- Accuracy
- Precision
- Recall
- F1-score

2.4.9 Hybrid Recommendation Score

The final recommendation score combines multiple evidence sources:

Final Score = $0.45 \times \text{Similarity Score} + 0.20 \times \text{Rule Score} + 0.20 \times \text{Random Forest Probability} + 0.15 \times \text{Feedback Score}$

This score balances semantic similarity, mechanical safety, learned compatibility, and real-world user feedback.

The weighting ensures that no single factor dominates the recommendation decision.

2.4.10 Explanation Builder

An explanation builder is included to improve user trust. Each recommendation includes:

- Why recommended
- Matched fields
- Key differences
- Substitute level
- Confidence score
- Evidence source

Example explanation:

“The part is recommended because it belongs to the same machine family and compatibility group, has the same function type, and has only a minor diameter difference.”

This helps users understand whether the part is an exact substitute, near substitute, or functional substitute.

2.4.11 API and Frontend Integration

The recommender is integrated through backend APIs. The workflow is:

1. User selects or searches a spare part.
2. Backend retrieves candidate parts.
3. Hard rules filter invalid candidates.
4. Similarity ranking scores valid candidates.
5. Random Forest predicts compatibility probability.
6. Hybrid score ranks candidates.
7. Explanation builder generates user-readable reasons.
8. Recommendations are returned to the frontend.

The final output includes:

- Recommended part list
- Compatibility score
- Substitute level
- Comparison details
- Explanation
- Feedback option

2.5 Component D: Blockchain-Based Spare Parts Verification and Traceability System

2.5.1 Methodological Approach

The blockchain component follows a **system design and implementation methodology** supported by field-level requirement analysis. Its purpose is to ensure spare part authenticity, traceability, ownership tracking, and tamper-proof lifecycle recording.

This component functions as the trust layer of the Smart Farmer platform. It uses blockchain to prevent counterfeit parts and maintain transparent ownership records.

2.5.2 Field Study and Requirement Analysis

A field study was conducted with:

- Farmers
- Spare part sellers
- Buyers
- Workshop technicians

The purpose was to understand how spare parts are currently purchased, verified, transferred, and recorded.

The study identified several issues:

- Farmers rely heavily on trust.
- Counterfeit parts are difficult to identify.
- Sellers do not use standardized digital records.
- Ownership transfers are informal.
- Buyers cannot verify product origin.
- Lifecycle records are incomplete or unavailable.

These findings shaped the blockchain system requirements.

2.5.3 System Requirements

The following requirements were identified:

- Unique digital identity for each spare part
- Secure QR-based verification
- Tamper-proof transaction records
- Ownership transfer tracking
- User-friendly access for non-technical users
- Hybrid storage for performance
- Backend API integration
- Cloud deployment

2.5.4 Blockchain Architecture Design

The system uses a hybrid architecture consisting of:

- Application layer
- Backend layer
- Blockchain layer
- Off-chain database layer

The blockchain layer stores critical immutable records such as registration, verification, and ownership transactions. The PostgreSQL database stores metadata and supports efficient retrieval.

This hybrid approach balances security and performance.

2.5.5 Hyperledger Fabric Network Setup

Hyperledger Fabric was selected because it is a permissioned blockchain framework. It allows controlled access, making it suitable for supply chain systems.

The network setup includes:

- Peer nodes
- Orderer node
- Channels
- Certificate authority
- Chaincode
- Docker containers
- AWS EC2 deployment

Docker was used to containerize blockchain components, while AWS EC2 provided cloud hosting for real-world accessibility.

2.5.6 Smart Contract Design

Smart contracts were developed to automate key blockchain operations.

The major smart contract functions include:

RegisterPart

Creates a new spare part record on the blockchain. It stores the unique part ID, owner details, registration time, and transaction hash.

VerifyPart

Checks whether a scanned or submitted part ID exists on the blockchain and confirms authenticity.

TransferOwnership

Records ownership transfer from one stakeholder to another.

The use of smart contracts ensures that all operations follow predefined rules and cannot be altered manually.

2.5.7 QR-Based Authentication

Each registered spare part is linked to a QR code. The QR code contains:

- Unique spare part ID
- Secure token
- Verification endpoint or reference

When the QR code is scanned:

1. The system reads the part ID and token.
2. Backend validates the token.
3. Blockchain is queried.
4. Authenticity result is returned.
5. User sees verification status.

This allows non-technical users to verify parts quickly using a mobile device.

2.5.8 Ownership Transfer Mechanism

Ownership transfer is implemented as a controlled two-step process:

1. Buyer initiates a transfer request.
2. Current owner approves the transfer.
3. Blockchain records the transfer transaction.
4. Database updates the current owner.
5. Previous ownership history remains preserved.

This prevents unauthorized transfers and creates a complete lifecycle history of the part.

2.5.9 Backend Integration

FastAPI is used as the backend framework. The backend handles:

- User requests
- QR token generation
- Data validation
- Blockchain communication
- PostgreSQL interaction
- API responses

The backend acts as the bridge between the user interface and the blockchain network.

2.5.10 System Deployment

The blockchain network and backend are deployed on AWS EC2. Cloud deployment allows:

- Remote access
- Scalability
- Multi-user testing
- Real-world simulation

The system is designed for integration with the Smart Farmer mobile/web application.

2.5.11 System Evaluation

The blockchain component is evaluated using:

- Functional correctness
- Transaction response time
- Verification success rate
- Tamper resistance
- Ownership transfer reliability
- Security and access control

The individual blockchain component reported approximate response times of:

- Registration: 2–5 seconds
- Verification: 1–2 seconds
- Ownership transfer: 3–6 seconds

2.6 Overall System Integration Methodology

2.6.1 Integration Approach

After developing each component independently, the modules are integrated into a unified Smart Farmer platform.

The integration workflow is:

1. User captures or uploads a spare part image.
2. Image Recognition module identifies the spare part.
3. 3D Visualization module highlights the identified part.
4. Wear Detection module detects defects and calculates severity.
5. Compatibility Recommendation module suggests suitable replacements if needed.
6. Blockchain module verifies authenticity of selected spare parts.
7. Final output is displayed to the user through the mobile/web interface.

2.6.2 Data Flow Between Components

The system uses structured JSON responses to exchange data between modules.

Example data flow:

- Image Recognition Output:
 - Part ID
 - Part name
 - Confidence score
- Wear Detection Output:
 - Defect class
 - Bounding box coordinates
 - Severity score
- Recommendation Output:
 - Compatible part ID
 - Substitute level
 - Compatibility score
 - Explanation
- Blockchain Output:

- Authenticity status
- Ownership history
- Verification timestamp

2.6.3 Backend and API Layer

The backend layer coordinates all modules. It handles:

- Image upload
- Model inference requests
- Database queries
- Recommendation generation
- Blockchain verification
- User feedback
- Response formatting

A modular API design is used so that each component can be updated independently without affecting the entire system.

2.6.4 Database Layer

The database stores:

- Spare part details
- Image classification labels
- 3D model mappings
- Compatibility schema records
- User feedback
- Off-chain blockchain metadata
- Recommendation history

This provides a centralized operational database while blockchain handles immutable trust records.

2.6.5 User Interface Integration

The frontend is designed for farmers, vendors, and mechanics. It provides:

- Image upload/capture

- Spare part identification results
- 3D visualization
- Wear detection results
- Compatible part recommendations
- QR verification
- Ownership transfer interface
- Maintenance-related outputs

The interface emphasizes simplicity because many users may not have advanced technical knowledge.

2.7 Testing and Evaluation Methodology

The system is evaluated at both component level and integrated system level.

2.7.1 Component-Level Testing

Each component is tested separately:

- Image Recognition: classification accuracy, confusion matrix
- Wear Detection: mAP50, precision, recall, F1-score
- Recommendation: Precision@5, Recall@5, classifier accuracy
- Blockchain: verification success, transaction response time, tamper resistance

2.7.2 Integration Testing

Integration testing checks whether modules communicate correctly.

Test cases include:

- Image recognition output correctly loads 3D model
- Wear detection output correctly generates severity data
- Recommendation module uses correct part data
- Blockchain verifies recommended parts
- Frontend displays combined output correctly

2.7.3 Usability Testing

Usability testing focuses on whether farmers and non-technical users can operate the system easily.

Evaluation areas include:

- Ease of image upload
- Clarity of detection results
- Usefulness of 3D visualization
- Understandability of recommendation explanations
- Ease of QR verification

2.7.4 Security Testing

Security testing is applied mainly to blockchain and backend components.

It includes:

- QR token validation
- Unauthorized transfer prevention
- API input validation
- Blockchain immutability verification
- Access control testing

CHAPTER 3: RESULTS AND DISCUSSION

3.1 System Implementation and Functional Results

3.1.1 System Deployment and Operational Overview

The proposed Blockchain-Based Spare Parts Verification and Traceability System was successfully implemented and deployed using a combination of Hyperledger Fabric, FastAPI backend services, PostgreSQL database, and QR-based authentication mechanisms. The blockchain network was deployed on an AWS EC2 instance, ensuring real-world accessibility and scalability, while backend services were hosted in a cloud environment to facilitate API communication.

The system was evaluated through multiple functional scenarios, including spare part registration, QR-based verification, and ownership transfer processes. Each component was tested individually and as part of the integrated system workflow to ensure reliability and correctness.

The deployment environment closely simulates real-world conditions, allowing the system to be assessed not only for correctness but also for usability and performance under realistic constraints.

3.1.2 Spare Part Registration and Blockchain Recording

The first major functionality evaluated was the spare part registration process. When a new spare part is registered, the system performs two key operations:

1. Metadata storage in the PostgreSQL database
2. Transaction recording on the blockchain via smart contract invocation

The system successfully generates a unique identifier for each part and records its details on the blockchain. The blockchain transaction hash is returned and stored as a reference for future verification.

From testing, it was observed that:

- All registered parts were successfully recorded on the blockchain
- Each transaction produced a unique hash, ensuring traceability
- Data stored on-chain remained immutable across all test cases

These results confirm that the blockchain layer effectively ensures data integrity and supports the requirement for tamper-proof record keeping, as emphasized in prior research on blockchain-based supply chains [3], [4].

3.1.3 QR-Based Verification Results

The QR-based verification mechanism was evaluated by scanning generated QR codes associated with registered spare parts. Each QR code contains a secure JWT token linking the physical part to its blockchain record.

The verification workflow was tested under multiple scenarios:

- Valid QR codes
- Tampered QR tokens
- Invalid or unregistered serial numbers

The results showed that:

- Valid QR codes consistently returned correct authenticity data
- Tampered tokens were successfully detected and rejected
- Invalid serial numbers resulted in appropriate error responses

This demonstrates that the QR-based authentication mechanism provides a secure and efficient method for verifying spare part authenticity. The combination of cryptographic token validation and blockchain querying ensures both security and reliability.

3.1.4 Ownership Transfer Workflow Evaluation

The ownership transfer functionality was tested using a two-step process:

1. Buyer initiates transfer request
2. Current owner approves the request

Upon approval, the system updates both the blockchain record and the database.

The evaluation revealed:

- Ownership transfers were successfully recorded on the blockchain
- Previous ownership data remained preserved, ensuring traceability
- Unauthorized transfer attempts were prevented

This confirms that the system effectively implements a secure ownership management mechanism, addressing the lack of transaction transparency observed in traditional supply chains [6].

3.2 Performance Evaluation

3.2.1 Transaction Response Time Analysis

System performance was evaluated by measuring response times for key operations:

- Blockchain registration
- Verification queries
- Ownership transfer

It was observed that:

- Registration transactions took approximately 2–5 seconds
- Verification queries were completed within 1–2 seconds
- Ownership transfers required 3–6 seconds due to multi-step validation

These response times are acceptable for the intended use case, where real-time processing is not critical but reliability is essential.

3.2.2 System Scalability Considerations

The deployment on AWS EC2 allows the system to scale based on demand. The use of Docker containers for blockchain components further enhances scalability and flexibility.

The hybrid architecture (on-chain + off-chain storage) also contributes to performance optimization, as large datasets are not stored directly on the blockchain.

3.2.3 Reliability and System Stability

Throughout testing, the system demonstrated consistent behavior with no data loss or transaction failures. Blockchain immutability ensured that once transactions were recorded, they remained permanently accessible.

3.3 Security Analysis

3.3.1 Data Integrity and Immutability

One of the key strengths of the system is its ability to ensure data integrity through blockchain immutability. Once a spare part is registered, its data cannot be modified or deleted.

This directly addresses the issue of data manipulation found in centralized systems [3].

3.3.2 Authentication and Access Control

Security mechanisms implemented include:

- JWT-based QR authentication
- Backend validation
- Permissioned blockchain access (Hyperledger Fabric)

These measures ensure that only authorized users can interact with the system.

3.3.3 Resistance to Counterfeit and Fraud

The system effectively prevents counterfeit spare parts by:

- Linking each part to a unique blockchain record
- Enabling real-time verification
- Detecting tampered QR codes

This aligns with research highlighting blockchain as a solution for counterfeit prevention [1], [4].

3.4 Discussion

3.4.1 Effectiveness of Blockchain-Based Approach

The results demonstrate that blockchain provides a highly effective solution for spare part verification and traceability. Compared to traditional systems, the proposed approach offers:

- Enhanced transparency
- Improved trust among stakeholders
- Secure and immutable data storage

These advantages confirm the suitability of blockchain technology for supply chain applications [8].

3.4.2 Importance of Field Study Integration

The inclusion of field study insights played a critical role in system design. Features such as QR-based verification and simplified workflows were directly influenced by real-world observations.

This highlights the importance of combining technical solutions with practical user requirements.

3.4.3 System Limitations

Despite its effectiveness, the system has some limitations:

- Dependence on internet connectivity
- Limited testing scope (local deployment environment)

- Lack of large-scale user testing

These limitations provide directions for future improvements.

3.4.4 Practical Impact and Usability

The system demonstrates strong potential for real-world adoption, particularly in agricultural environments where counterfeit spare parts are a major issue.

The QR-based interface ensures ease of use, even for non-technical users, making the system accessible and practical.

CHAPTER 4: CONCLUSION AND FUTURE WORK

4.1. Summary of Findings

This report has presented the design, development, and evaluation of the Blockchain-Based Spare Part Verification and Traceability Module, which functions as the trust and security layer of the Smart Spare Parts Management System. The research was motivated by the significant operational, economic, and safety challenges associated with counterfeit spare parts and lack of traceability in agricultural machinery, particularly in smallholder farming environments where maintenance decisions are often reactive and based on limited information.

The research followed a structured system design and implementation approach, integrating field-level requirement analysis with blockchain-based architecture development. Unlike traditional centralized systems, which are vulnerable to data manipulation and lack transparency, the proposed system leverages blockchain technology to provide a decentralized, immutable, and verifiable record of spare part lifecycle events.

A key contribution of this research lies in the incorporation of real-world field insights into system design. Through interactions with farmers, spare part sellers, and buyers, it was observed that current spare part transactions are predominantly informal, lacking any reliable verification or ownership tracking mechanisms. Farmers rely heavily on trust and prior experience when purchasing spare parts, while sellers operate without standardized systems for recording transactions or verifying authenticity. These findings confirmed that the absence of a structured and secure system is a major factor contributing to the widespread presence of counterfeit components and inefficient maintenance practices.

The system implementation was evaluated across multiple functional scenarios, including spare part registration, QR-based verification, and ownership transfer workflows. The results demonstrate that the blockchain-based approach effectively addresses the identified challenges.

The first major outcome of the research is the successful implementation of a **blockchain-based registration system**, where each spare part is assigned a unique digital identity and recorded on the blockchain ledger. This ensures that once a part is registered, its data remains immutable and cannot be altered or deleted. The use of smart contracts enables automated enforcement of system rules, ensuring consistency and reliability in transaction processing.

The second key finding relates to the effectiveness of the **QR-based verification mechanism**. By linking each physical spare part to a secure digital record through a QR code containing a cryptographically signed token, the system enables fast and reliable authenticity checks. Testing results showed that valid QR codes consistently returned accurate data, while tampered or invalid codes were successfully detected and rejected. This demonstrates that the integration of QR-based authentication with blockchain querying provides a practical and secure solution for real-world verification.

The third significant outcome is the implementation of a **controlled ownership transfer mechanism**. Unlike traditional systems where ownership changes are undocumented, the proposed system records each transfer as a blockchain transaction. The two-step approval process ensures that ownership changes occur only with mutual consent, thereby preventing unauthorized transfers. Additionally, the complete ownership history of each part is preserved, enabling full lifecycle traceability.

Another important contribution of this research is the adoption of a **hybrid system architecture**, combining blockchain and off-chain storage. While critical data such as ownership and verification records are stored on the blockchain, metadata is maintained in a PostgreSQL database. This approach balances performance and security, ensuring efficient data retrieval without compromising data integrity.

From a performance perspective, the system demonstrated acceptable response times for all major operations. Blockchain transactions, including registration and ownership transfer, were completed within a few seconds, while verification operations were performed almost instantly. These results indicate that the system is suitable for real-world deployment, where reliability is more critical than ultra-low latency.

Security analysis further confirms the effectiveness of the proposed approach. The use of blockchain ensures data immutability, while JWT-based QR authentication prevents unauthorized access and duplication. The permissioned nature of Hyperledger Fabric provides controlled access, ensuring that only authorized participants can interact with the system.

Overall, the research demonstrates that a blockchain-based approach is not only technically feasible but also highly effective in addressing the challenges of spare part verification and traceability. The integration of real-world field insights with advanced technologies results in a system that is both practical and scalable.

4.2 Future Work

4.2.1 Expansion to Additional Spare Part Categories

The current implementation focuses on a generalized spare part management framework; however, future work can extend the system to support specific categories of spare parts with customized attributes and lifecycle models. For example:

- Engine components (filters, pistons, belts)
- Brake systems (pads, drums)
- Hydraulic components (hoses, valves)
- Transmission parts (chains, gears)

Each category may require specialized metadata, verification criteria, and lifecycle tracking models. Expanding the system to cover these categories would significantly increase its applicability and impact across different sectors.

4.2.2 Integration with IoT-Based Monitoring Systems

A major limitation of the current system is its reliance on user-initiated actions, such as scanning QR codes. Future work can integrate the system with IoT sensors to enable automated condition monitoring.

For example:

- Vibration sensors to detect abnormal machine behavior
- Temperature sensors to monitor overheating components
- Pressure sensors for hydraulic systems

Sensor data can be used to trigger verification requests or maintenance alerts, enabling a shift from reactive to predictive maintenance.

4.2.3 Advanced Ownership and Supply Chain Analytics

While the current system provides basic ownership tracking, future enhancements can introduce advanced analytics capabilities, including:

- Supply chain flow analysis
- Identification of counterfeit distribution patterns
- Usage trend analysis for different spare parts
- Predictive demand forecasting

These features would transform the system from a verification tool into a comprehensive supply chain intelligence platform.

4.2.4 Scalability and Large-Scale Deployment

The current implementation has been tested in a controlled environment. Future work should focus on large-scale deployment involving multiple stakeholders, including manufacturers, distributors, and service providers.

This would require:

- Optimization of blockchain performance
- Implementation of load balancing mechanisms
- Enhanced network configuration for scalability

4.2.5 User Interface and Mobile Application Development

Although the backend system is fully functional, the user interface component remains limited. Future work should focus on developing a user-friendly mobile application that simplifies interaction with the system.

Features may include:

- One-click QR scanning
- Visual verification results
- Ownership transfer notifications
- Maintenance reminders

A well-designed interface is critical for adoption among non-technical users such as farmers.

4.2.6 Integration with Government and Regulatory Systems

Another potential extension is the integration of the system with regulatory authorities to enforce standards and prevent counterfeit distribution.

This could include:

- Certification of genuine spare parts
- Monitoring of supply chain activities
- Enforcement of compliance regulations

Such integration would enhance trust and ensure wider adoption of the system.

4.2.7 Continuous System Improvement through Feedback

As the system is deployed in real-world environments, user feedback can be used to improve functionality and performance. Future work can include:

- Feedback-driven feature enhancements
- System optimization based on usage patterns
- Continuous security improvements

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